

Multi-Cafe POS / Cafe Application – Simple Schema Document

This document explains a **professional yet simple database structure** for a **multi-café POS application**. It supports both **quick billing cafés** and **table-based cafés**, with GST, UI control, permissions, and printing.

1. Core Concept

- One application
 - Multiple cafés (companies)
 - Each café has its own users, items, billing rules, UI, and print settings
 - All data is separated using `company_id`
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2. Company (Cafe) Master

companies

Stores café-level configuration.

Key fields: - `company_name` - `has_gst` (true / false) - `enable_tables` (true / false) - `print_enabled` - `print_format` (58mm / 80mm / A4)

Purpose: - Controls GST usage - Controls whether table-based orders are used - Controls printing behavior

3. GST & HSN (Optional)

Used only if `companies.has_gst = true`

hsn_master

- `company_id`
- `hsn_code`
- `gst_percentage`

If GST is disabled, these tables are ignored.

4. Users, Roles & Permissions

Roles

- ADMIN (1 per company)
- MANAGER (1 per company)
- USER (2 or more)

users

- company_id
- role_id
- name
- mobile
- password

permissions

Defines actions like: - CREATE_BILL - VIEW_REPORT - MANAGE_ITEMS

role_permissions

Maps roles to permissions.

Purpose: - Show / hide UI features - Secure actions per role

5. UI Configuration

ui_themes

Predefined UI styles: - CLASSIC - MODERN - QUICK_BILL

company_ui_settings

- default_ui_theme
- allow_user_ui_change

user_ui_preferences

- user_id
- selected_ui_theme

Purpose: - User selects UI on first login - Café owner can lock or allow UI change

6. Item Group Logic (Tea Example)

item_groups

- group_name (Tea, Coffee)

- display_type (GROUPED / DIRECT)
- show_separate_items (true / false)

Logic: - DIRECT → Clicking Tea adds directly to bill - GROUPED + true → Tea opens item list - GROUPED + false → Only one Tea button

This supports cafés that bill only "Tea" without variants.

7. Items

items

- company_id
- item_group_id
- item_name
- rate
- taxable (true / false)
- hsn_id (optional)

Items are visible only for their café.

8. Table Management (Optional)

Enabled only if `companies.enable_tables = true`

cafe_tables

- table_name (T1, T2)
- seating_capacity

table_status

- table_id
- status (FREE / OCCUPIED)

If tables are disabled, this module is ignored completely.

9. Orders (Live Running Orders)

Order is different from Bill.

orders

- company_id
- table_id (nullable)
- order_type (DINE_IN / TAKEAWAY)
- status (OPEN / CLOSED)

order_items

- order_id
- item_id
- qty

Purpose: - Take multiple orders before billing - Used mainly for table-based cafés

10. Billing (Final Settlement)

bills

- company_id
- order_id (optional)
- subtotal
- tax
- discount
- total
- payment_mode

bill_items

- bill_id
- item_id
- qty
- rate

Quick cafés can directly create bills without orders.

11. Printing Configuration

print_settings

- company_id
- printer_type
- paper_size
- show_logo
- show_gst
- footer_text

Controlled fully by café owner.

12. Application Flow

Without Tables (Tea Shop)

1. Login
2. Add items

3. Generate bill
4. Print

With Tables (Restaurant)

1. Login
 2. Select table
 3. Create order
 4. Add items
 5. Generate bill
 6. Print & free table
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13. Why This Design is Professional

- Multi-company safe
 - Optional complexity
 - Role-based access
 - Flexible item grouping
 - GST optional
 - Ready for future expansion (KOT, split bill, reports)
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14. Recommended Backend

- PostgreSQL + Supabase
 - SQLite for offline (sync later)
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